

Statistical Models in Computational Biology

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Due date: 22nd April 2026

Problem 19: Exponential family of the Negative Binomial distribution (3 points)

The Negative Binomial distribution plays a fundamental role in biological research for modeling count data (like RNA-seq reads or microbial populations). It can be thought of as a more flexible version of the Poisson distribution as it accounts for overdispersion, where natural variation causes the variance to be much higher than the mean. Knowing that it is in the Exponential Family allows us to use the Negative Binomial distribution with standard GLM techniques.

The probability mass function (PMF) of the Negative Binomial distribution is given by:

$$f(x|p) = \binom{x+r-1}{x} p^x (1-p)^r \quad (1)$$

where p is the success probability, and r is the number of failures.

For the rest of the exercise assume r is a known constant (i.e., we are only interested in p).

(a) Proof that the Negative Binomial distribution belongs to the Exponential family and define all of the components as defined in the lecture notes (η, τ, h, α) . (1 points)

(b) Derive the expected value of a Negative Binomial random variable and re-write the Exponential family formulation in terms of the expected value $\eta := \mu = E[X]$. (2 points)

Problem 20: Sparsity and metabolic engineering in *B. subtilis* (3 points)

Solve this exercise in R using the `glmnet` package.

The file `riboflavin.csv` contains genomic data for the bacteria *Bacillus subtilis*. The first column, `y`, represents the log-production rate of riboflavin (Vitamin B2), and the remaining 4,088 columns correspond to gene expression levels for a total of $n = 71$ observations. Your task is to identify a sparse set of genomic predictors for vitamin production.

(a) **Data Preparation:** Load the data. Construct the response vector $\mathbf{y}$ and the covariate matrix $\mathbf{X}$. Randomly split the dataset into a training set (70%) and a test set (30%). For reproducibility, set a seed at the beginning of your script. (0.5 points)

(b) **Model Fitting:** Using 10-fold cross-validation on the training set, find the optimal penalty parameter λ (using `lambda.min`) for the following three models:

- Lasso regression ($\alpha = 1$)
- Ridge regression ($\alpha = 0$)

- Elastic Net regression ($\alpha = 0.5$)

(1 point)

(c) **Sparsity and Performance:** For each of the three models:

- Report the number of non-zero coefficients (excluding the intercept) selected at `lambda.min`.
- Predict the riboflavin production on the test set and calculate the Mean Squared Error (MSE).

(1.5 points)

Problem 21: Identification of protein-protein interaction networks via the graphical lasso (4 points)

Solve this exercise again in R. Use the `glmnet` packages for model fitting and the accompanying script `plotcirculargraph.R` for the final figure.

You are given protein expression data in human immune system cells. Precisely, you are given data `proteinexpression.csv`, containing 7466 cells and flow cytometry measurements of 11 phosphorylated proteins and phospholipids. Our goal here is to identify the interaction network between the different proteins. Your task is to implement the approximate version of the graphical lasso (as describe in the script) to estimate the corresponding interaction network and visualize it.

- Load `proteinexpression.csv`
- Loop over all proteins (each protein as the response once with all others as the predictors) and fit a lasso model (cross-validated on 5 folds, select parameters corresponding to `lambda.1se` for penalization strength) (2 point)
- Build the interaction matrix between all proteins. Define no interaction as 0 and an interaction as 1. (An interaction is found if both lasso regressions showed a non-zero effect for a measurement pair) (1 point)
- Set row and column names for the interaction matrix and use the given function to plot the network. (1 point)